Real-World Cyber Attack Case Studies

Virus, Malware and Ransomware Attacks on Servers, Corporations and Government Bodies

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Introduction

This report studies five well-documented cyber incidents in which viruses, ransomware or malware damaged servers, disrupted organisations, or caused data loss and theft. Three cases are from outside India (UK, global shipping, USA) and two are from India. For each case the report gives the timing, cause, loss, recovery, cost and the solutions adopted, with links to the original news articles and official sources as proof.

Note on figures: losses are estimates reported by the organisations, governments or news outlets cited in each case, and sources sometimes differ slightly.

Summary of the five cases

Case	Year	Type	Reported loss
WannaCry / NHS	2017	Ransomware worm	GBP 92 million (NHS)
NotPetya / Maersk	2017	Wiper malware	USD 250-300 million (Maersk)
Colonial Pipeline	2021	Ransomware	USD 4.4 million ransom; fuel shortages
AIIMS Delhi	2022	Ransomware	About a week of manual operation
Cosmos Bank, Pune	2018	Malware / SWIFT fraud	Rs 94.42 crore

Case 1: WannaCry Ransomware Attack on the UK NHS and the World (2017)

Victim	UK National Health Service (NHS) plus 200,000+ computers in about 150 countries
Date / timing	12 May 2017 (a Friday); main disruption 12-18 May 2017
Type of attack	Ransomware worm using the EternalBlue Windows (SMB) vulnerability
Main loss	Estimated GBP 92 million to the NHS; about 19,000 appointments cancelled

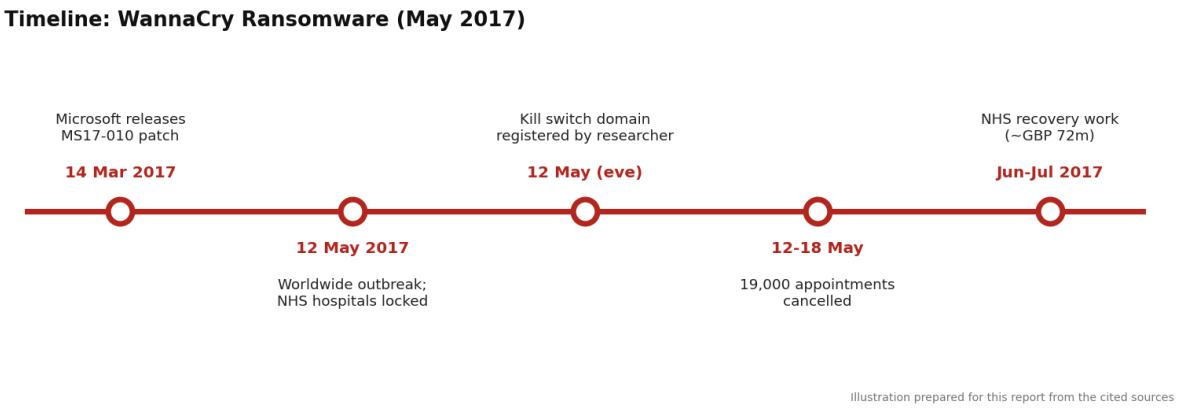


Figure: Incident timeline (illustration prepared for this report from the cited sources)

What happened

On 12 May 2017 the WannaCry worm spread automatically between unpatched Windows computers and encrypted their files, showing a red screen demanding payment in Bitcoin. In England it disrupted about one-third of hospital trusts and around 8% of GP practices. Some hospitals had to divert ambulances and cancel operations and appointments.

Causes

- Many NHS machines still ran outdated Windows versions or had not installed Microsoft's March 2017 security patch (MS17-010).
- Weak network segmentation let the worm spread freely between systems.
- The National Audit Office described it as a relatively unsophisticated attack that basic IT security practice could have prevented.

Loss and damage

- Department of Health and Social Care estimate: about GBP 92 million in total.
- About GBP 19 million was lost patient-care output during the attack week; about GBP 72 million went on IT recovery work in June-July 2017.
- About 19,000 appointments cancelled; globally, over 200,000 computers affected.

Recovery and cost of recovery

- A security researcher registered an unregistered 'kill switch' domain hidden in the malware, which stopped the spread on the same day.

- Affected machines were cleaned and restored from backups, and patches were applied.
- The UK government pledged an extra GBP 150 million over three years for NHS cyber security. Most victims did not need to pay, as WannaCry was widely reported to have no reliable way of returning files after payment.

Solution and lessons

- Apply security patches promptly and retire unsupported operating systems.
- Segment networks and keep offline backups.
- Run a national Cyber Security Operations Centre and regular security testing.

Sources (original articles / proof)

- https://www.nationalhealthexecutive.com/articles/wannacry-cyber-attack-cost-nhs-ps92m-after-19000-appointments-were-cancelled
- https://www.healthcareitnews.com/news/nhs-department-health-warned-get-their-act-together-after-national-audit-office-wannacry
- https://www.publicfinance.co.uk/node/29254

Newspaper coverage — Case 1

Newspaper: The Guardian | Article date: 27 October 2017



Figure 1: NHS could have avoided WannaCry hack with 'basic IT security', says report

Authentic online newspaper article screenshot; not a scan of the print edition.

Source: https://www.theguardian.com/technology/2017/oct/27/nhs-could-have-avoided-wannacry-hack-basic-it-security-national-audit-office

Case 2: NotPetya Attack on Maersk, the World's Largest Shipping Company (2017)

Victim	A.P. Moller-Maersk (shipping and ports), plus Merck, FedEx/TNT and others
Date / timing	27 June 2017; recovery took about 10 days, with disruption until mid-August 2017
Type of attack	Destructive wiper disguised as ransomware (NotPetya), spread through a poisoned software update
Main loss	About USD 250-300 million for Maersk; over USD 10 billion globally (widely cited estimate)

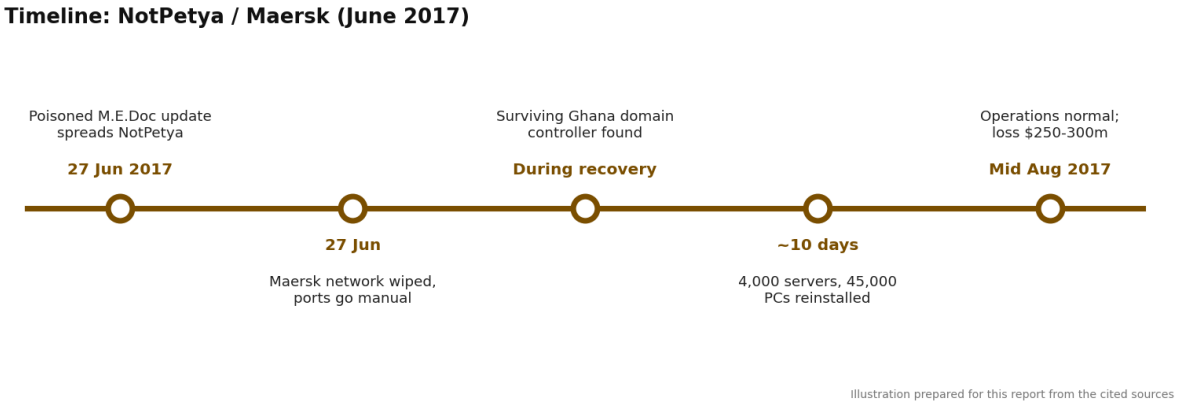


Figure: Incident timeline (illustration prepared for this report from the cited sources)

What happened

NotPetya was planted in updates of M.E.Doc, a Ukrainian accounting software. It then spread across company networks using the EternalBlue exploit. At Maersk it wiped the entire Windows network, shutting down booking systems and port terminals around the world.

Causes

- Compromised update server of a third-party software supplier (a supply-chain attack).
- Unpatched Windows systems and a flat internal network let the malware spread quickly.
- Maersk's chairman later admitted the company had been only 'average' in cyber security before the attack.

Loss and damage

- Maersk: USD 250-300 million in direct losses.
- 4,000 servers, 45,000 PCs and 2,500 applications had to be reinstalled.
- Business ran largely manually; a volume drop of about 20% was reported during the outage. Merck lost about USD 870 million and FedEx/TNT about USD 400 million.

Recovery and cost of recovery

- A single domain controller in Ghana, which had been offline because of a power cut, survived and held a clean copy of the network's directory data.

- IT teams rebuilt the entire infrastructure in about 10 days, a job normally estimated at around six months.
- Customers diverted some bookings during the first two weeks, then volumes recovered.

Solution and lessons

- Keep offline, tested backups of critical directory services.
- Segment networks and patch quickly.
- Vet third-party software suppliers; Maersk aimed to turn cyber security into a competitive advantage.

Sources (original articles / proof)

- https://itpro.com/cyber-attacks/30393/maersk-rebuilt-hefty-it-infrastructure-a-mere-10-days-after-notpetya-attack
- https://scworld.com/news/notpetya-attack-totally-destroyed-maersks-computer-network-chairman
- https://www.supplychaindive.com/news/maersk-reconstructed-entire-it-infrastructure-in-10-days-following-nyetya/515776/

Newspaper coverage — Case 2

Newspaper: The Guardian | Article date: 27 June 2017



Figure 2: 'Petya' ransomware attack strikes companies across Europe and US

Authentic online newspaper article screenshot; not a scan of the print edition.

Source: https://www.theguardian.com/world/2017/jun/27/petya-ransomware-attack-strikes-companies-across-europe
This contemporary report uses the name “Petya”; the incident is subsequently identified as NotPetya and includes Maersk among the affected companies.

Case 3: Colonial Pipeline Ransomware Attack, USA (2021)

Victim	Colonial Pipeline Company, the largest refined-fuel pipeline in the USA (about 5,500 miles)
Date / timing	7 May 2021; pipeline restarted 12 May 2021
Type of attack	DarkSide ransomware (ransomware-as-a-service group)
Main loss	USD 4.4 million ransom paid (75 bitcoin); about 5 days of shutdown and fuel shortages

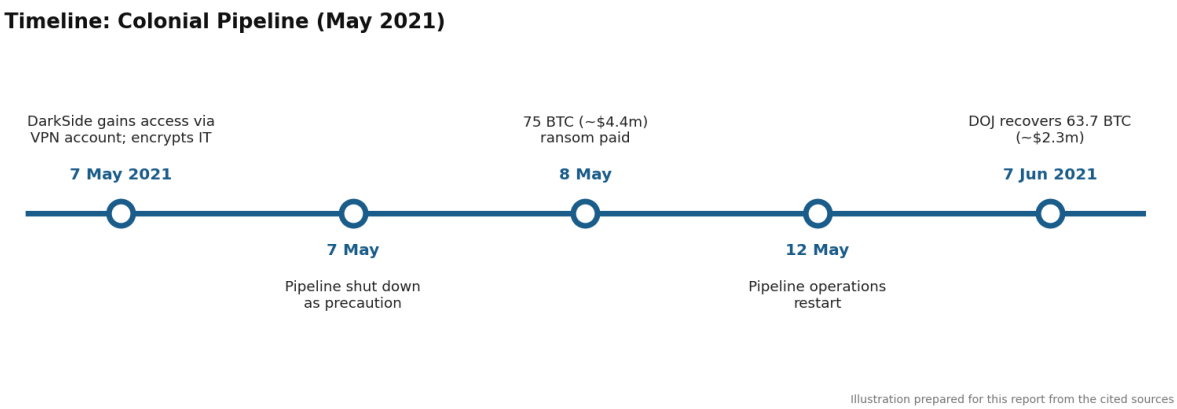


Figure: Incident timeline (illustration prepared for this report from the cited sources)

What happened

Attackers entered Colonial's network and encrypted IT systems on 7 May 2021. As a precaution the company shut down the pipeline, which supplies a large share of the fuel on the US East Coast. Panic buying followed and some petrol stations ran dry.

Causes

- Access came through a legacy VPN account whose password had probably been exposed elsewhere, according to the incident responders at Mandiant.
- The account was reportedly not protected by multi-factor authentication.

Loss and damage

- Ransom: about 75 bitcoin (about USD 4.4 million at the time).
- Several days of halted fuel transport and public panic buying.
- Large incident-response and recovery costs.

Recovery and cost of recovery

- Colonial paid the ransom and received a decryption tool, but restoring systems was still slow, so operations restarted on 12 May.
- On 7 June 2021 the US Department of Justice announced it had seized 63.7 of the 75 bitcoin (about USD 2.3 million), tracing it to a wallet whose private key the FBI held.
- The DOJ credited Colonial's quick notification of law enforcement.

Solution and lessons

- Enforce multi-factor authentication and remove unused VPN accounts.
- Separate IT and operational (pipeline) networks.
- Report incidents to law enforcement quickly; maintain tested incident-response plans.

Sources (original articles / proof)

- https://www.techtarget.com/whatis/feature/Colonial-Pipeline-hack-explained-Everything-you-need-to-know
- https://www.theregister.com/2021/06/08/antiransomware_task_force/
- https://www.businessinsurance.com/doj-recovers-most-of-ransom-paid-by-colonial-pipeline/

Newspaper coverage — Case 3

Newspaper: The Guardian | Article date: 20 May 2021

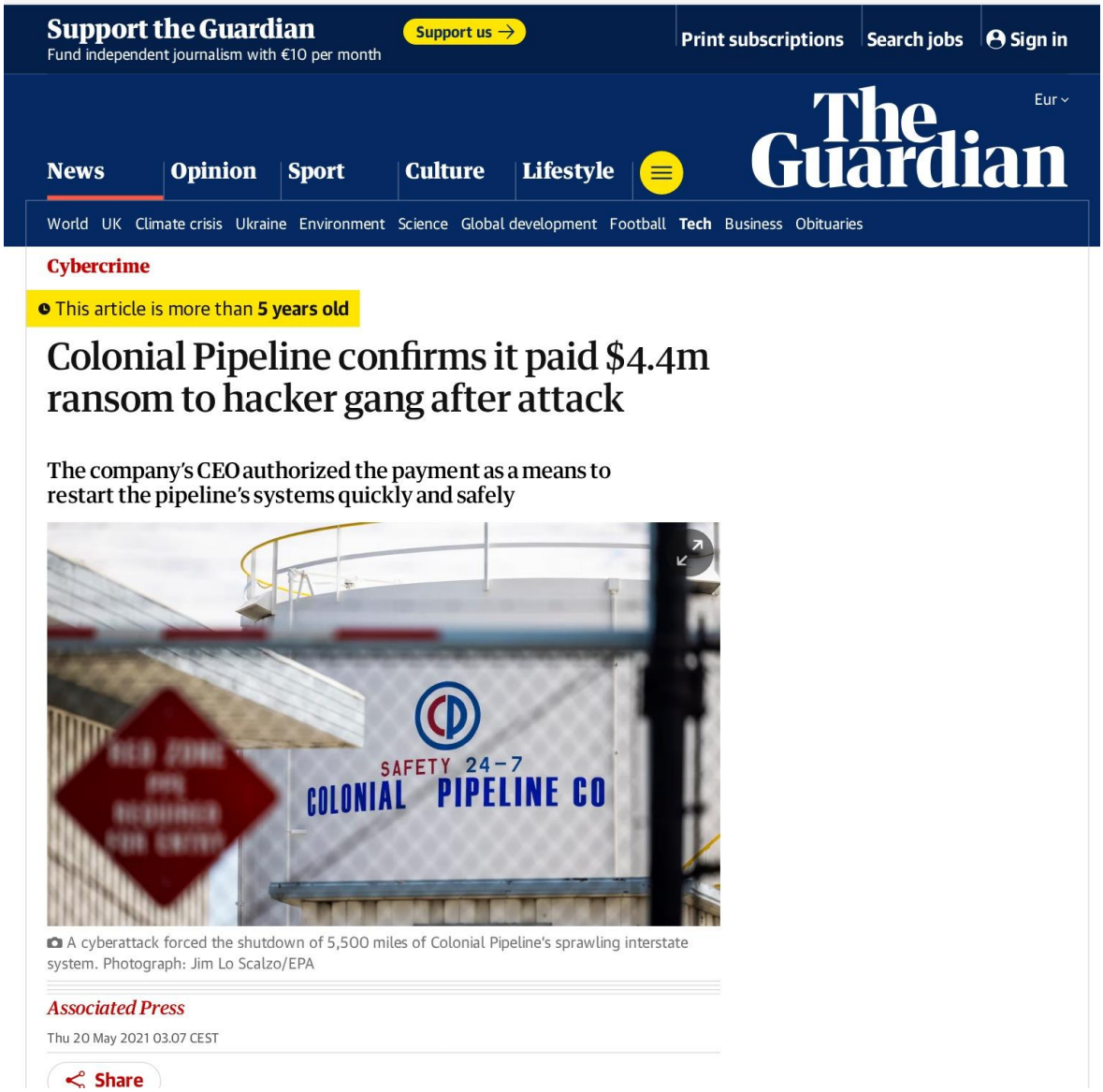


Figure 3: Colonial Pipeline confirms it paid $4.4m ransom to hacker gang after attack

Authentic online newspaper article screenshot; not a scan of the print edition.

Source: https://www.theguardian.com/technology/2021/may/19/colonial-pipeline-cyber-attack-ransom
Date shown above follows the article's displayed publication date, rather than the date embedded in its URL.

Case 4: AIIMS Delhi Ransomware Attack, India (2022)

Victim	All India Institute of Medical Sciences (AIIMS), New Delhi - a major government hospital
Date / timing	First signs about 6:45-7:00 am on 23 November 2022; data restored 29 November 2022
Type of attack	Suspected ransomware: files changed extensions and primary and backup servers were corrupted
Main loss	About a week with all digital services down; hospital run manually

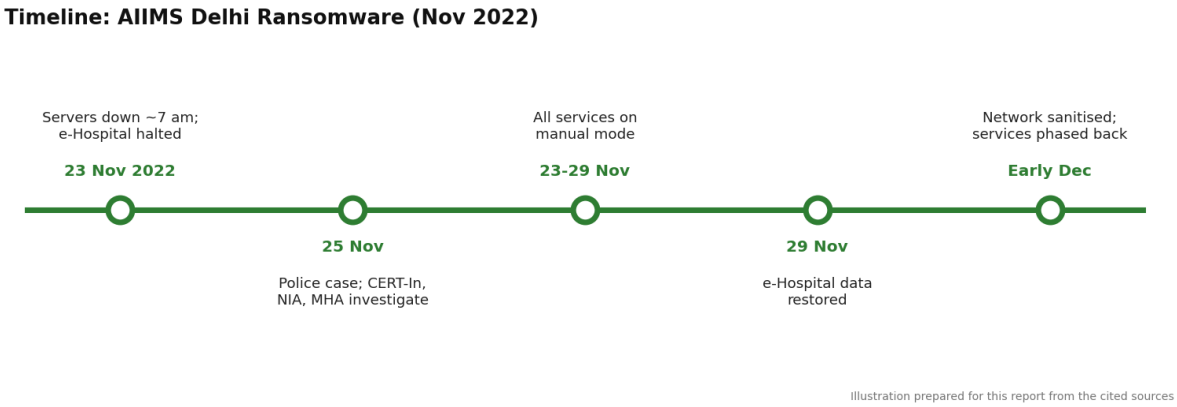


Figure: Incident timeline (illustration prepared for this report from the cited sources)

What happened

AIIMS' e-Hospital servers run by the National Informatics Centre stopped working on 23 November 2022. Outpatient and inpatient digital services, smart lab, billing, report generation and the appointment system were affected. Staff handled admissions, discharges, lab work and records manually, and patients faced long delays for reports.

Causes

- Ransomware that encrypted files on the main and backup servers.
- Delhi Police registered a case of extortion and cyber terrorism; CERT-In, the Home Ministry and other agencies investigated.
- Media reports claimed a demand of about Rs 200 crore in cryptocurrency, but Delhi Police said no ransom demand had been brought to their notice by AIIMS.

Loss and damage

- Services running in manual mode for roughly a week or more; test reports delayed for thousands of patients.
- The server held data of many VIP patients, raising data-privacy concerns.
- No official figure for the financial cost was published.

Recovery and cost of recovery

- AIIMS stated that the e-Hospital data was restored on 29 November 2022, and extra physical servers were arranged for restoration.

- The network was sanitised and antivirus tools were deployed on servers and computers before services returned in phases.
- No ransom payment was officially confirmed.

Solution and lessons

- Isolate and sanitise the entire network before bringing services back.
- Maintain separate, tested backups and an offline copy.
- Strengthen monitoring and cyber-security audits for government hospitals.

Sources (original articles / proof)

- https://scroll.in/latest/1038702/how-aiims-is-working-to-restore-services-after-hackers-crippled-its-computer-systems
- https://www.deccanherald.com/amp/story/india%2Faiims-server-down-various-agencies-looking-into-incident-delhi-police-file-fir-1165511.html
- https://www.businesstoday.in/amp/latest/trends/story/aiims-delhi-says-e-hospital-data-been-restored-on-servers-services-continue-to-run-on-manual-mode-354643-2022-11-29

Newspaper coverage — Case 4

Newspaper: Hindustan Times | Article date: 23 November 2022 (updated)

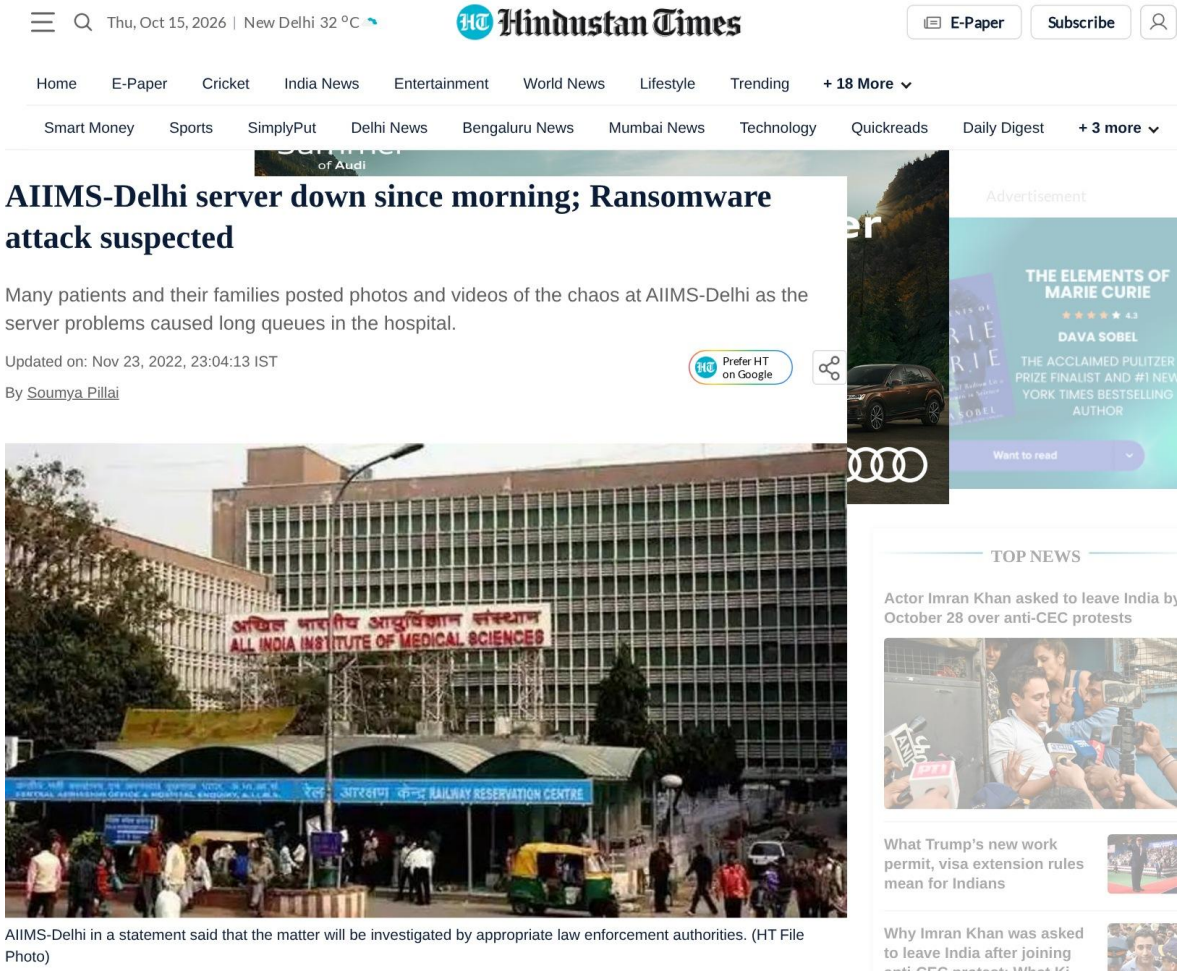


Figure 4: AIIMS-Delhi server down since morning; Ransomware attack suspected

Authentic online newspaper article screenshot; not a scan of the print edition.

Source: https://www.hindustantimes.com/cities/delhi-news/aiimsdelhi-server-down-since-morning-ransomware-attack-suspected-101669217988046.html

Case 5: Cosmos Bank Malware Attack, Pune, India (2018)

Victim	Cosmos Co-operative Bank Ltd, Pune (one of India's oldest and largest co-operative banks)
Date / timing	11 August 2018 (about 3 pm to 10 pm) and 13 August 2018
Type of attack	Malware on the ATM switch server plus fraudulent SWIFT transfers
Main loss	Rs 94.42 crore (about USD 13.5 million)

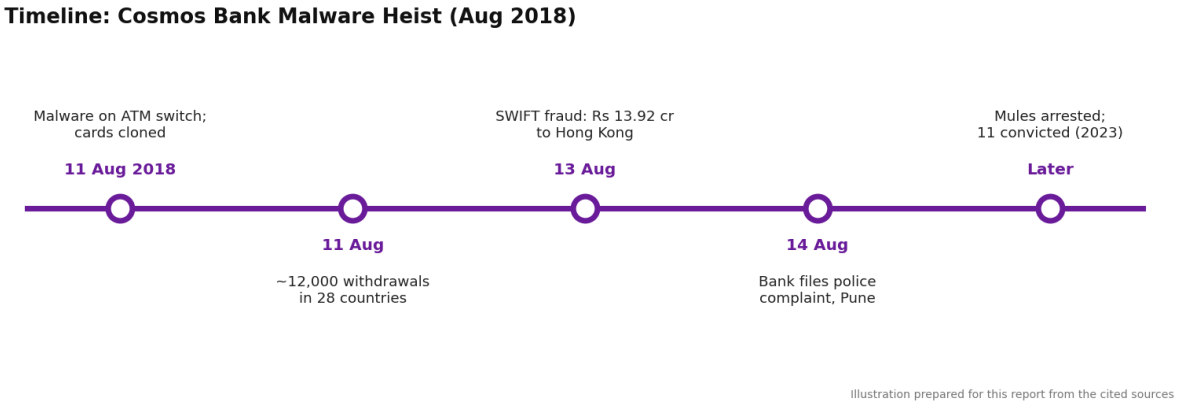


Figure: Incident timeline (illustration prepared for this report from the cited sources)

What happened

Hackers planted malware on the bank's ATM switch server and created a proxy switch that approved fraudulent card payments. Cloned Visa and RuPay debit cards were used for withdrawals in 28 countries, mostly in small amounts. On 13 August, Rs 13.92 crore was transferred through SWIFT to a Hong Kong account.

Causes

- Malware attack on the bank's own IT environment (NPCI said its own systems were secure).
- The core banking system was not affected, so customer account balances stayed intact.
- The attack was reported to have originated from Canada, likely used as a relay.

Loss and damage

- Rs 78 crore withdrawn from ATMs abroad, Rs 2.5 crore within India, and Rs 13.92 crore via SWIFT.
- About 12,000 transactions abroad plus further domestic withdrawals.

Recovery and cost of recovery

- The bank filed a police complaint, brought in a forensic agency and reconciled card transactions with Visa and RuPay.
- At least 18 people, mainly money mules, were arrested across India.
- In 2023 a Pune court convicted 11 people (nine got four years, two got three years).

Solution and lessons

- Isolate payment switches and monitor for proxy or rogue switches.

- Real-time transaction and SWIFT monitoring and independent security audits.
- Faster reconciliation and alerts with card networks and law enforcement.

Sources (original articles / proof)

- https://moneylife.in/article/npci-blames-cosmos-bank-for-malware-attacks/55026.html
- https://www.businesstoday.in/amp/industry/banks/story/pune-cosmos-bank-atm-server-hacked-rs-94-crore-transferred-to-overseas-accounts-109361-2018-08-14
- https://business-standard.com/amp/topic/cosmos-bank-fraud

Newspaper coverage — Case 5

Newspaper: The Times of India | Article date: 15 August 2018

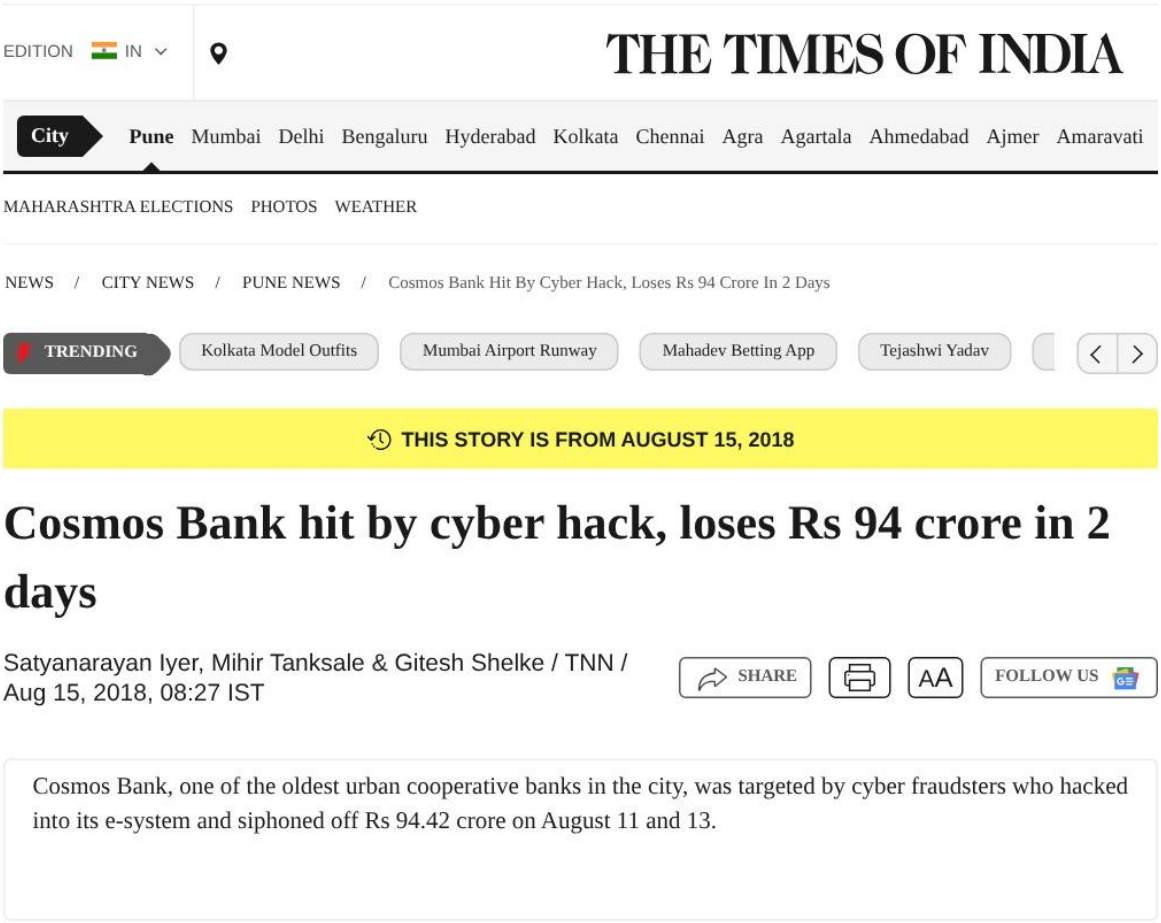


Figure 5: Cosmos Bank hit by cyber hack, loses Rs 94 crore in 2 days

Authentic online newspaper article screenshot; not a scan of the print edition.

Source: https://timesofindia.indiatimes.com/city/pune/cosmos-bank-hit-by-cyber-hack-loses-rs-94-crore-in-2-days/articleshow/65409441.cms

Conclusion

These cases show that most major incidents start with a small weakness: an unpatched system, a reused password, a poisoned software update or an unsegmented network. Organisations that kept tested backups, involved law enforcement early and patched quickly recovered faster. The main defences are regular patching, multi-factor authentication, network segmentation, offline backups, staff training and a rehearsed incident-response plan.